📘 Backend Development

🎯 Course Description

This course teaches students to **build, secure, and maintain backend systems** for applications. It covers **server-side programming, API development, security, and deployment** to create scalable and reliable backend applications.

🎓 Learning Outcomes

By the end of this course, students will be able to:

1. Define backend development and its role in applications
2. Build server-side applications using programming languages like Node.js, Python, or Java
3. Create RESTful APIs and integrate with frontend systems
4. Implement authentication, authorization, and security best practices
5. Debug, troubleshoot, and optimize backend code
6. Deploy backend applications to cloud or server environments
7. Apply version control (Git) for backend projects

🧩 Course Structure

🟢 MODULE 1: INTRODUCTION TO BACKEND DEVELOPMENT

Topics:

- What is backend development
- Backend vs frontend
- Common backend technologies: Node.js, Python (Flask/Django), Java (Spring Boot)
- Web servers: Apache, Nginx

Practical:

- Install Node.js/Python and run a basic “Hello World” server

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🟡 MODULE 2: SERVER-SIDE PROGRAMMING

Topics:

- Handling HTTP requests and responses
- Routing, controllers, and middleware
- JSON and data processing

Practical:

- Build a basic REST API endpoint
- Test API with Postman

🔵 MODULE 3: API DEVELOPMENT & INTEGRATION

Topics:

- RESTful API principles
- CRUD operations (Create, Read, Update, Delete)
- Consuming external APIs

Practical:

- Build an API for a task management app
- Fetch data from a public API

🟣 MODULE 4: BACKEND SECURITY

Topics:

- Authentication: sessions, tokens, JWT
- Authorization and role-based access
- Password hashing & encryption
- Preventing vulnerabilities: SQL Injection, XSS, CSRF

Practical:

- Implement JWT authentication
- Hash passwords with bcrypt

🔴 MODULE 5: DEBUGGING, LOGGING & PERFORMANCE

Topics:

- Error handling and logging
- Debugging techniques
- Query and code optimization
- Monitoring performance

Practical:

- Log API requests and errors
- Optimize slow queries

🟠 MODULE 6: DEPLOYMENT & VERSION CONTROL

Topics:

- Git: commit, push, pull, branching
- Deploy backend apps on cloud (Heroku, AWS, Azure)
- CI/CD overview

Practical:

- Push project to GitHub
- Deploy API to Heroku

🧪 Final Assessment

- Build a complete backend application:
  - RESTful API
  - Authentication & authorization
  - Integration with database (optional for assessment)
- Present architecture, security, and optimization

📊 Assessment Breakdown

- Assignments → 20%

- Practical Work → 40%
- Final Project → 40%